Simulacro Parcial 1

Cálculo Integral

Departamento de
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Nombre: _____ **Cédula** _____

La siguiente lista de antiderivadas puede ser empleada para la solución del simulacro

- $\int dx = x + C.$
- $\int x^n dx = \frac{x^{n+1}}{n+1} + C$, si $n \neq -1$.
- $\int \frac{1}{x} dx = \ln|x| + C.$
- $\int e^x dx = e^x + C.$
- $\int a^x dx = \frac{a^x}{\ln(a)} + C$, si $a > 0$, con $a \neq 1$.
- $\int \text{sen}(x) dx = -\cos(x) + C.$
- $\int \cos(x) dx = \text{sen}(x) + C.$
- $\int \sec^2(x)\, dx = \tan(x) + C.$
- $\int \sec(x)\tan(x)\, dx = \sec(x) + C.$
- $\int \csc^2(x)\, dx = -\cot(x) + C.$
- $\int \csc(x)\cot(x)\, dx = -\csc(x) + C.$
- $\int \frac{1}{\sqrt{1-x^2}}\, dx = \text{sen}^{-1}(x) + C.$
- $\int \frac{1}{1+x^2}\, dx = \tan^{-1}(x) + C.$

I. Selección

En las preguntas 1 a 4 seleccione la respuesta adecuada.

1. [10 Puntos] Sea f una función continua definida en el intervalo $[0,4]$ tal que $\int_0^2 f(t)\, dt = 4$ y $\int_2^4 f(t)\, dt = 6$. El valor de de la integral $\int_0^2 f(2t)\, dt$ es:

(a) 4 (b) 5 (c) 6 (d) 10 (e) Ninguna de las anteriores

$u = 2t$

$du = 2dt$

$dt = \frac{du}{2}$

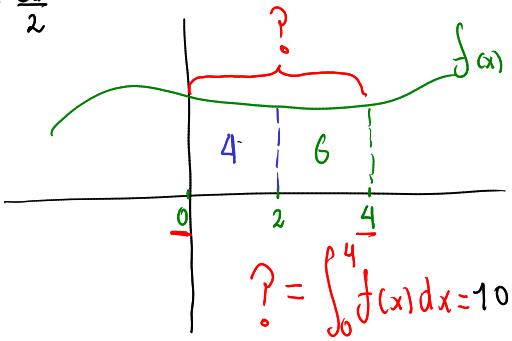
Reemplazar $\int_0^2 dt \quad u = 2t$

$t = 2 \to u = 2(2) = 4$

$t = 0 \to u = 2(0) = 0$

$\int_0^4 f(u)\frac{du}{2} \to \frac{1}{2}\int_0^4 f(u)\, du$

$\to \frac{1}{2}(10) = 5$ ✓



2. [10pt] Suponga que g es una función continua definida en el intervalo $[0,4]$ tal que $\int_0^4 g(x)dx = 8$. El valor de $\int_0^2 (g(2x)-1)dx$ es:

$$\int_0^2 g(2x)-1\,dx \rightarrow \int_0^2 g(2x)dx - \int_0^2 1\,dx \rightarrow 4-2=2$$

$1[2-0]$

$u=2x$

$du=2dx$

$dx=\frac{du}{2}$

$2\rightarrow x=2 \quad u=4$

$dx \quad u=2x$

$0\rightarrow x=0 \rightarrow u=0$

$$\int_0^4 g(u)\frac{du}{2} \rightarrow \frac{1}{2}\int_0^4 g(u)du \rightarrow \frac{1}{2}(8)=4$$

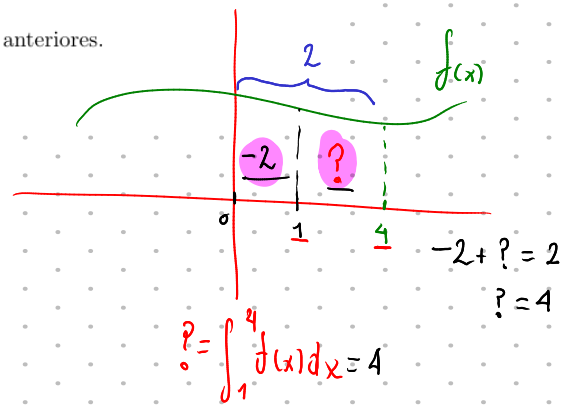
2. [10pt] Sea f una función continua definida en el intervalo $[0,4]$ tal que $\int_0^1 f(x)dx = -2$ y $\int_0^4 f(x)dx = 2$.

El valor de $\int_1^4 (3f(x)-2)dx$ es

(a) 6, (b) 10, (c) -2, (d) 2, (e) Ninguna de las anteriores.

$$3\int_1^4 f(x)dx - \int_1^4 2\,dx \rightarrow 3(4)-6=6$$

$2[4-1]$



$-2+?=2$

$?=4$

$$?=\int_1^4 f(x)dx = 4$$

2. [10 Puntos] Desde el fondo de un tanque sale agua a razón de $v(t) = 60 - 4t$ litros por minuto, donde $t \in [0, 15]$. La cantidad de agua que sale del tanque durante los primeros 10 minutos es:

(a) 200 litros (b) 300 litros (c) 400 litros (d) 500 litros (e) Ninguna de las anteriores

Velocidad $V(t) = 60 - 4t$ L/min

$$\#Agua = \int_0^{10} 60 - 4t \, dt \rightarrow 60t - \frac{4t^2}{2} \Big|_0^{10} \rightarrow 60t - 2t^2 \Big|_0^{10} \rightarrow 60(10) - 2(10)^2$$
$$= 600 - 200 = 400 \, Litros$$

3. [10 Puntos] Considere la función definida por $F(x) = \int_{x^2}^{4} \frac{2^t}{t} \, dt$. Entonces, el valor de $F'(2)$ es:

(a) -32 (b) -64 (c) -8 (d) -16 (e) Ninguna de las anteriores

4. [10 Puntos] El valor de la integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\text{sen}(x) + \cos(x))^2 \, dx$ es igual a:

(a) 2π (b) 4π (c) $\frac{\pi}{2}$ (d) π (e) Ninguna de las anteriores

$$(Sen(x) + Cos(x))^2 = Sen^2(x) + 2\,Sen(x)\cdot Cos(x) + Cos^2(x) = \underbrace{Sen^2(x) + Cos^2(x)}_{1} + \underbrace{2\,Sen(x)\cdot Cos(x)}_{Sen(2x)}$$

$$\int_{-\pi/2}^{\pi/2} 1 + Sen(2x)\,dx = x - \frac{Cos(2x)}{2} \Big|_{-\pi/2}^{\pi/2} \checkmark\checkmark$$

$$\int Sen(2x)\,dx = -\frac{Cos(2x)}{2}$$

$u = 2x$, $du = 2\,dx$, $dx = \frac{du}{2}$

$$\int Sen(u)\frac{du}{2} \rightarrow \frac{1}{2}\int Sen(u)\,du = -\frac{Cos(u)}{2}$$

$$\int Sen(ax)\,dx = -\frac{Cos(ax)}{a} + C$$

$$\int Cos(ax)\,dx = \frac{Sen(ax)}{a} + C$$

$\int Sen^2\theta$ $\int Cos^2\theta$

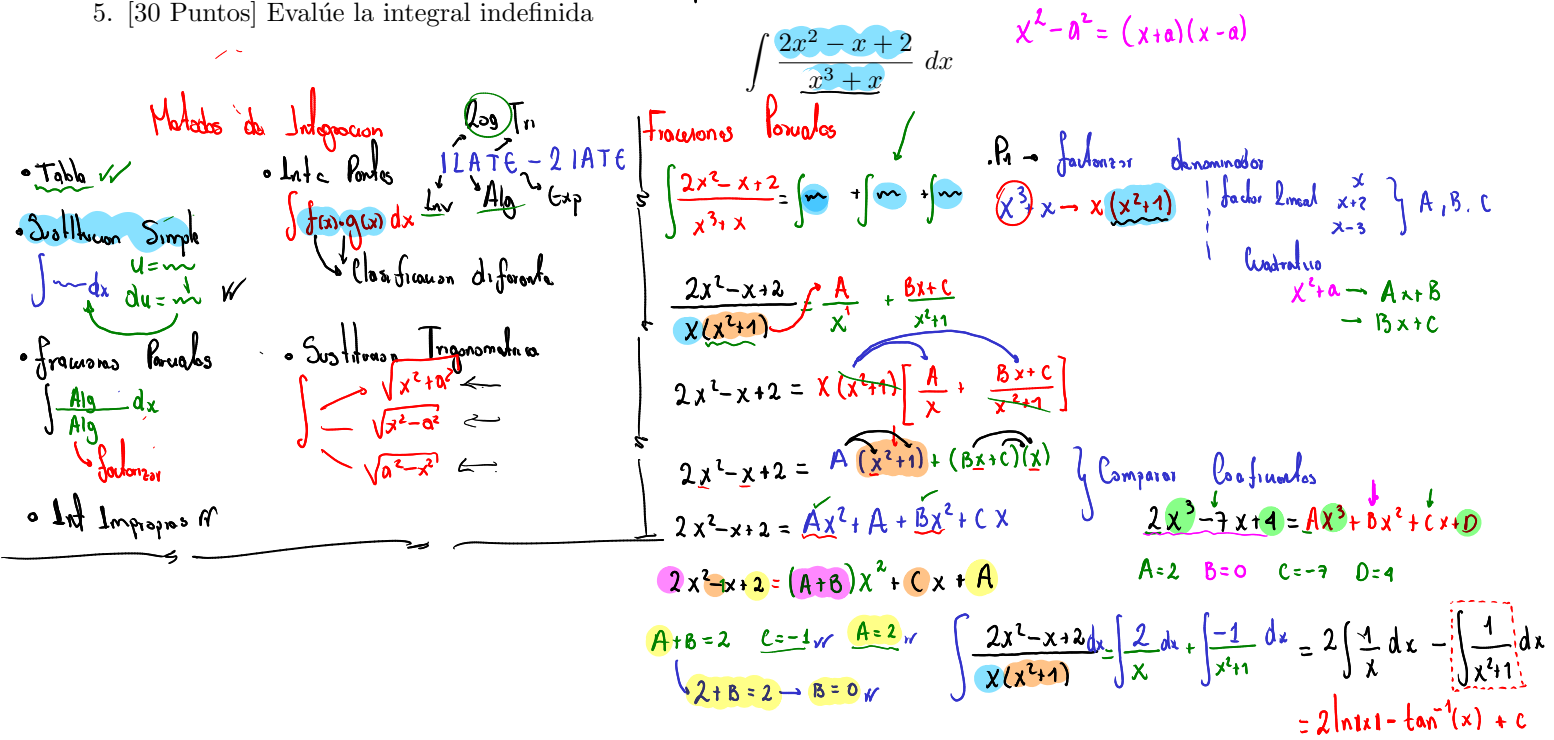
Básicas y Suma/Resta	Ángulo Doble y Medio	Suma a Producto; Producto a Suma
$\sin\theta = \frac{1}{\csc\theta}$ ✓	$\sin(2\theta) = 2\sin\theta\cos\theta$ ✓✓	$\sin\theta + \sin\beta = 2\sin\left(\frac{\theta+\beta}{2}\right)\cos\left(\frac{\theta-\beta}{2}\right)$
$\cos\theta = \frac{1}{\sec\theta}$ ✓	$\cos(2\theta) = \cos^2\theta - \sin^2\theta$	$\sin\theta - \sin\beta = 2\sin\left(\frac{\theta-\beta}{2}\right)\cos\left(\frac{\theta+\beta}{2}\right)$
$\csc\theta = \frac{1}{\sin\theta}$ ✓	$\cos(2\theta) = 1 - 2\sin^2\theta$	$\cos\theta + \cos\beta = 2\cos\left(\frac{\theta+\beta}{2}\right)\cos\left(\frac{\theta-\beta}{2}\right)$
$\sec\theta = \frac{1}{\cos\theta}$ ✓	$\cos(2\theta) = 2\cos^2\theta - 1$	$\cos\theta - \cos\beta = -2\sin\left(\frac{\theta+\beta}{2}\right)\sin\left(\frac{\theta-\beta}{2}\right)$
$\cot\theta = \frac{\cos\theta}{\sin\theta} = \frac{1}{\tan\theta}$ ✓	$\tan(2\theta) = \frac{2\tan\theta}{1-\tan^2\theta}$	$\sin\theta\sin\beta = \frac{1}{2}[\cos(\theta-\beta) - \cos(\theta+\beta)]$
$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1}{\cot\theta}$ ✓	$\sin\frac{\theta}{2} = \pm\sqrt{\frac{1-\cos\theta}{2}}$	$\cos\theta\cos\beta = \frac{1}{2}[\cos(\theta-\beta) + \cos(\theta+\beta)]$
$\sin^2\theta + \cos^2\theta = 1$ ✓✓	$\cos\frac{\theta}{2} = \pm\sqrt{\frac{1+\cos\theta}{2}}$	$\sin\theta\cos\beta = \frac{1}{2}[\sin(\theta+\beta) + \sin(\theta-\beta)]$
$1 + \tan^2\theta = \sec^2\theta$ ✓✓	$\tan\frac{\theta}{2} = \pm\sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$	**Suplemento, complemento**
$1 + \cot^2\theta = \csc^2\theta$	$\tan\frac{\theta}{2} = \frac{1-\cos\theta}{\sin\theta} = \frac{\sin\theta}{1+\cos\theta}$	$\sin(\pi \pm \theta) = \mp\sin\theta$
$\sin(\theta \pm \beta) = \sin\theta\cos\beta \pm \sin\beta\cos\theta$	**Par, impar**	$\cos(\pi \pm \theta) = -\cos\theta$
$\cos(\theta \pm \beta) = \cos\theta\cos\beta \mp \sin\theta\sin\beta$	$\sin(-\theta) = -\sin\theta$	$\sin(\pi/2 - \theta) = \cos\theta$
$\tan(\theta \pm \beta) = \frac{\tan\theta \pm \tan\beta}{1 \mp \tan\theta\tan\beta}$	$\cos(-\theta) = \cos\theta$	$\cos(\pi/2 - \theta) = \sin\theta$

II. Solución con procedimiento

En las preguntas 5 y 6 se debe incluir el procedimiento necesario para determinar y justificar sus respuestas. Respuestas sin procedimiento no serán tenidas en cuenta.

5. [30 Puntos] Evalúe la integral indefinida

$$\int \frac{2x^2 - x + 2}{x^3 + x}\, dx$$



6. [30 Puntos] Calcule la integral indefinida

$$\int x\tan^{-1}(x)\, dx.$$

1. $\int u\,dv = uv - \int v\,du$
2. $\int u^n\,du = \frac{u^{n+1}}{n+1} + C$
3. $\int \frac{du}{u} = \ln u + C$
4. $\int e^u\,du = e^u + C$
5. $\int a^u\,du = \frac{a^u}{\ln a} + C$
6. $\int \sin u\,du = -\cos u + C$
7. $\int \cos u\,du = \sin u + C$
8. $\int \sec^2 u\,du = \tan u + C$
9. $\int \csc^2 u\,du = -\cot u + C$
10. $\int \sec u\tan u\,du = \sec u + C$
11. $\int \csc u\cot u\,du = -\csc u + C$
12. $\int \tan u\,du = \ln|\sec u| + C$
13. $\int \cot u\,du = \ln|\sin u| + C$
14. $\int \sec u\,du = \ln|\sec u + \tan u| + C$
15. $\int \csc u\,du = \ln|\csc u - \cot u| + C$
16. $\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1}\left(\frac{u}{a}\right) + C$
17. $\int \frac{du}{a^2 + u^2} = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right) + C$
18. $\int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a}\sec^{-1}\left(\frac{u}{a}\right) + C$
19. $\int \frac{du}{a^2 - u^2} = \frac{1}{2a}\ln\left|\frac{u+a}{u-a}\right| + C$
20. $\int \frac{du}{u^2 - a^2} = \frac{1}{2a}\ln\left|\frac{u-a}{u+a}\right| + C$

$$\int \frac{1}{x^2+1}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)$$

6. [16pt] Calcular la integral indefinida $\int \frac{1}{x^2+3x}dx$.

$$\int \frac{1}{x^2+3x}\,dx$$

$$\frac{1}{x^2+3x} = \frac{1}{x(x+3)} = \frac{A}{x} + \frac{B}{x+3}$$

$$1 = x(x+3)\left[\frac{A}{x} + \frac{B}{x+3}\right]$$

Factores lineales

$$1 = A(x+3) + B(x)$$

$x=-3$ $x=0$

$x=0 \rightarrow 1 = A(3) \rightarrow A = 1/3$

$x=-3 \rightarrow 1 = B(-3) \rightarrow B = -1/3$

$$\int \frac{1}{x(x+3)}\,dx = \int \frac{1/3}{x}\,dx + \int \frac{-1/3}{x+3}\,dx$$

$$= \frac{1}{3}\int \frac{1}{x}\,dx - \frac{1}{3}\int \frac{1}{x+3}\,dx$$

$$= \frac{1}{3}\ln|x| - \frac{1}{3}\ln|x+3| + C$$

$$\int \frac{1}{x+3}\,dx$$

$u = x+3$

$du = 1\,dx$

$dx = du$

Reemplazar $\int \frac{1}{u}\,du = \ln|u|$

Vol a la variable original

$\ln|x+3|$ √√

$$\left(\tan^{-1}(u)\right)' = \frac{u'}{u^2+1}$$

6. [30 Puntos] Calcule la integral indefinida

$$\int x\tan^{-1}(x)\,dx.$$

Inversa

Alg

Clasificar diferente

$\int f(x)\cdot g(x)\,dx$

I → Inversa

L → Logaritmicas

A → Alg

T → Trigonometricas

E → Exponenciales

ILATE

$\tan^{-1}(x)$ x

$u = \tan^{-1}(x)$ $\quad du = \frac{1}{x^2+1}dx$

$dv = x\,dx$ $\quad v = \frac{x^2}{2}$

$$\int u\,dv = uv - \int v\,du \rightarrow \tan^{-1}(x)\frac{x^2}{2} - \int \frac{x^2}{2}\cdot\left(\frac{1}{x^2+1}\right)dx$$

$$\frac{a+b}{c} = \frac{a}{c} + \frac{b}{c}$$

$$\int \frac{x^2}{2}\cdot\left(\frac{1}{x^2+1}\right)dx \rightarrow \frac{1}{2}\int \frac{x^2+1-1}{x^2+1}\,dx \rightarrow \frac{1}{2}\left[\int \frac{x^2+1}{x^2+1}\,dx - \int \frac{1}{x^2+1}\,dx\right]$$

$$\frac{1}{2}\left[\int 1\,dx - \tan^{-1}(x)\right] \rightarrow \frac{1}{2}\left[x - \tan^{-1}(x)\right]$$

$$\rightarrow \tan^{-1}(x)\frac{x^2}{2} - \int \frac{x^2}{2}\cdot\left(\frac{1}{x^2+1}\right)dx \rightarrow \frac{\tan^{-1}(x)\cdot x^2}{2} - \frac{1}{2}\left[x - \tan^{-1}(x)\right] + C \text{ √√}$$

1. $\frac{d}{dx}[cu] = cu'$
2. $\frac{d}{dx}[u \pm v] = u' \pm v'$
3. $\frac{d}{dx}[uv] = uv' + vu'$
4. $\frac{d}{dx}\left[\frac{u}{v}\right] = \frac{vu' - uv'}{v^2}$
5. $\frac{d}{dx}[c] = 0$
6. $\frac{d}{dx}[u^n] = nu^{n-1}u'$
7. $\frac{d}{dx}[x] = 1$
8. $\frac{d}{dx}[|u|] = \frac{u}{|u|}(u'),\ u \neq 0$
9. $\frac{d}{dx}[\ln u] = \frac{u'}{u}$
10. $\frac{d}{dx}[e^u] = e^u u'$
11. $\frac{d}{dx}[\log_a u] = \frac{u'}{(\ln a)u}$
12. $\frac{d}{dx}[a^u] = (\ln a)a^u u'$
13. $\frac{d}{dx}[\text{sen } u] = (\cos u)u'$
14. $\frac{d}{dx}[\cos u] = -(\text{sen } u)u'$
15. $\frac{d}{dx}[\tan u] = (\sec^2 u)u'$
16. $\frac{d}{dx}[\cot u] = -(\csc^2 u)u'$
17. $\frac{d}{dx}[\sec] = (\sec u \tan u)u'$
18. $\frac{d}{dx}[\csc u] = -(\csc u \cot u)u'$
19. $\frac{d}{dx}[\text{arcsen } u] = \frac{u'}{\sqrt{1-u^2}}$
20. $\frac{d}{dx}[\arccos u] = \frac{-u'}{\sqrt{1-u^2}}$
21. $\frac{d}{dx}[\arctan u] = \frac{u'}{1+u^2}$
22. $\frac{d}{dx}[\text{arccot } u] = \frac{-u'}{1+u^2}$
23. $\frac{d}{dx}[\text{arcsec } u] = \frac{u'}{|u|\sqrt{u^2-1}}$
24. $\frac{d}{dx}[\text{arccsc } u] = \frac{-u'}{|u|\sqrt{u^2-1}}$
25. $\frac{d}{dx}[\text{senh } u] = (\cosh u)u'$
26. $\frac{d}{dx}[\cosh] = (\text{senh } u)u'$
27. $\frac{d}{dx}[\tanh u] = (\text{sech}^2 u)u'$
28. $\frac{d}{dx}[\coth u] = -(\text{csch}^2 u)u'$
29. $\frac{d}{dx}[\text{sech } u] = -(\text{sech } u \tanh u)u$
30. $\frac{d}{dx}[\text{csch } u] = -(\text{csch } u \coth u)u'$
31. $\frac{d}{dx}[\text{senh}^{-1} u] = \frac{u'}{\sqrt{u^2+1}}$
32. $\frac{d}{dx}[\cosh^{-1} u] = \frac{u'}{\sqrt{u^2-1}}$
33. $\frac{d}{dx}[\tanh^{-1} u] = \frac{u'}{1-u^2}$
34. $\frac{d}{dx}[\coth^{-1} u] = \frac{u'}{1-u^2}$
35. $\frac{d}{dx}[\text{sech}^{-1} u] = \frac{-u'}{u\sqrt{1-u^2}}$
36. $\frac{d}{dx}[\text{csch}^{-1} u] = \frac{-u'}{|u|\sqrt{1+u^2}}$

4. [10pt] El valor de $\int_0^{\pi/2} x\cos(x)dx$ es:

(a) $\frac{\pi}{2}$, (b) $\frac{\pi}{2}+1$, (c) $\frac{\pi}{2}-1$, (d) 1, (e) Ninguna de las anteriores.

$\int_0^{\pi/2} x\cos(x)\,dx$

A T

Método de Tabulación

$\int A\cdot T \quad \int A\cdot\epsilon$

	Deri	Inte
+	x	$\cos(x)$
−	1	$\operatorname{Sen}(x)$
+	0	$-\operatorname{Cos}(x)$

$\int_0^{\pi/2} x\cos(x)\,dx = x\operatorname{Sen}(x)+\operatorname{Cos}(x)\Big|_0^{\pi/2} \to \frac{\pi}{2}\operatorname{Sen}(90)+\operatorname{Cos}(90)-(0\cdot\operatorname{Sen}(0)+\operatorname{Cos}(0))$

$\to \frac{\pi}{2}-1$

4. [10pt] El valor de $\int_0^1 2xe^x dx$ es:

(a) $2(e-1)$, (b) $2(e+1)$, (c) 2, (d) $2e$, (e) Ninguna de las anteriores.

$\int_0^1 2xe^x dx = 2xe^x - 2e^x\Big|_0^1$

A ϵ

Tabulación

	Der	Int
+	$2x$	e^x
−	2	e^x
+	0	e^x

3. [10pt] El valor de $\int_0^{\sqrt{\pi}} 3x\operatorname{sen}(x^2)dx$ es:

(a) 6, (b) 0, (c) $\frac{3}{2}$, (d) 3, (e) Ninguna de las anteriores.

$u=$ $du=$ ↳ Está en la Integral ! Sustitución → Reemplazar la variable

$\int 3x\operatorname{Sen}(x^2)dx$

↳ Reemplazar $\int 3x\operatorname{Sen}(u)\frac{du}{2x} \to \frac{3}{2}\int \operatorname{Sen}(u)du \to \frac{-3}{2}\operatorname{Cos}(u)$

$u=x^2$

$du=2xdx$

$dx=\frac{du}{2x}$

Volver a la V. Original

$\int_0^{\sqrt{\pi}} 3x\operatorname{Sen}(x^2)dx = \frac{-3}{2}\operatorname{Cos}(x^2)\Big|_0^{\sqrt{\pi}}$

2. El valor de $\int_0^1 t^5 e^{t^6}\,dt$ es:

(a) $\frac{e}{5}$, (b) $\frac{e}{6}$, (c) $\frac{e-1}{5}$, (d) $\frac{e-1}{6}$, (e) $e-1$.

$\int t^5 e^{t^6} dt$ → Reemplazo

$u = t^6$

$du = 6t^5 dt$

$dt = \frac{du}{6t^5}$

$\int t^5 e^u \frac{du}{6t^5} \rightarrow \frac{1}{6}\int e^u du \rightarrow \frac{1}{6} e^u$ ✓

Volver a la V.O

$\int_0^1 t^5 e^{t^6} dt = \frac{1}{6} e^{t^6}\Big|_0^1$ ✓✓

3. [10 Puntos] Considere la función definida por $F(x) = \int_{x^2}^{4} \frac{6t}{t}\,dt$. Entonces, el valor de $F'(2)$ es:

(a) -32 (b) -64 (c) -8 (d) -16 (e) Ninguna de las anteriores

- Derivar la Integral
- TFC 1

$f(x) = \int_{h(x)}^{g(x)} m(t)\,dt \rightsquigarrow f'(x) = g(x)'\cdot m(g(x)) - h'(x)\cdot m(h(x))$ ✓

$F'(x) = \frac{(4)'\cdot 2^{4}}{4} - \frac{(x^2)'\cdot 2^{x^2}}{x^2} \rightarrow F'(x) = -2x\cdot\frac{2^{x^2}}{x^2}$

$F'(2) = \frac{-2\cdot 2^4}{2} = -16$ ✓

3. [10pt] Sea $f(x) = \int_0^{\operatorname{sen}(x)} \sqrt{t^2+1}\,dt$. Entonces $f'(x)$ es igual a:

(a) $\cos(x)\sqrt{\operatorname{sen}^2(x)+1}$, ✓ (b) $\sqrt{\operatorname{sen}^2(x)+1}$, (c) $\cos(x)$,
(d) $-\cos(x)\sqrt{\operatorname{sen}^2(x)+1}$, (e) Ninguna de las anteriores.

$f'(x) = (\operatorname{Son}(x))'\sqrt{\operatorname{Son}^2(x)+1} - (0)'\sqrt{0^2+1}$

$= \operatorname{Cos}(x)\sqrt{\operatorname{Son}^2(x)+1}$ ✓✓